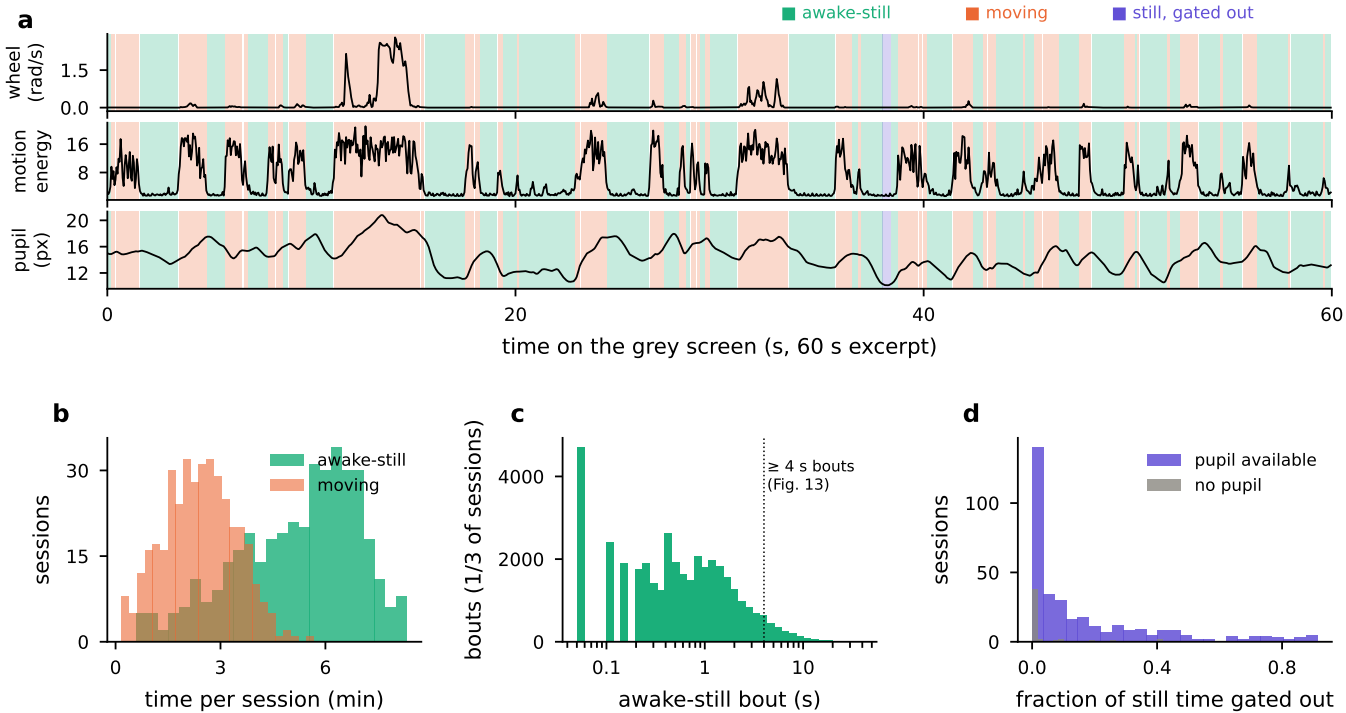

Figure 10: Defining awake stillness on the passive grey screen. **a** Example grey-screen minute (session 5d6aa933): wheel speed, left-camera motion energy and pupil diameter, with awake-still (green), moving (orange) and still-but-gated-out (violet) 50 ms bins. **b** Awake-still and moving time per session (389 sessions). **c** Durations of awake-still bouts (log axis); true-offset bouts of at least 4 s enter the within-bout analyses. **d** Fraction of still time removed by the anti-sleep gate (pupil below the 10th percentile of the task-period pupil, or more than 30 s since the last movement).